CECS 229 – Assignment 10

Determinants

The file, 'pa10.py', contains an incomplete set of functions that will calculate a determinant of a square matrix. Use the Matrix class that you wrote for Assignment 8, file 'pa8.py'.

Update the file 'pa10.py' to add functionality to calculate a determinant:

1. `submatrix(A, i, j)` – given a matrix A, row i, and column j, return the submatrix that is found by removing row i and column j from matrix A.
    a. A – a matrix.
    b. i – the index of the row that will be removed from A (1 based).
    c. j – the index of the column that will be removed from A (1 based).
    d. Returns a new Matrix that is the submatrix of A with row i and column j removed.
    e. Raise a IndexError if index i or j are not within the dimensions of A.
2. `determinant(A)` – given a square matrix, A, return the result of calculating the determinant of matrix A. Note: for matrices of size 1x1 or 2x2, use Leibniz Formula, for n>=3, use Laplace Expansion. Use the submatrix function that you wrote to find the submatrix when calculating the cofactors. Use recursion to find the determinants of these submatrices.
    a. A – a square matrix.
    b. Returns the value of the determinant.
    c. Raise a ValueError if matrix A is not square.

Example Output:

```
A = [[0, 2, 3, 1], [2, 4, 7, 3], [1, 1, 2, 2]]
submatrix(A, 2, 1) # ValueError
A = [[0, 2, 3, 1], [2, 4, 7, 3], [1, 1, 2, 2], [2, 1, 4, 3]]
submatrix(A, 2, 1) # [[2, 3, 1], [1, 2, 2], [1, 4, 3]]

A = [[0, 2, 3, 1], [2, 4, 7, 3], [1, 1, 2, 2]]
determinant(A) # ValueError
A = [[0, 2, 3, 1], [2, 4, 7, 3], [1, 1, 2, 2], [2, 1, 4, 3]]
determinant(A) # 6
```

Notes:

1. Please place your name in a comment block at the top of both 'pa8.py' and 'pa10.py'.
2. Fill in the necessary code to complete the functions in 'pa10.py'. Write your code where specified (ie. the TODOs). Do not modify the file names, function names, method names, or parameters otherwise the program and tests will not work. Return the results in the format specified (ie. correct types and container types), otherwise the program and tests will not work properly.
3. Your solutions may not use any libraries or functions that perform the equivalent task.
4. Test your code using the provided 'main.py' file. You can write additional tests in the 'your_tester.py' file to further check your solutions. Note: just because the main tests pass does not guarantee that it will pass the tests on Gradescope.
5. When you are finished, submit both your 'pa8.py' and 'pa10.py' files to Gradescope on Canvas for further testing and grading. It should allow you to make repeated submissions.